Curriculum | MSBA

MS in Business Analytics

The Tepper School of Business developed the curriculum for the online Master of Science in Business Analytics (MSBA) in consultation with global business leaders. The Tepper School determined that the greatest need is for professionals who not only have advanced analytical skills, such as machine learning and optimization, but also the appropriate business knowledge and communication skills to solve complex problems and bring value to industry.

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- Business domain knowledge, including marketing, operations, accounting, finance, and people analytics.



> **When I compared the Tepper MSBA with other similar programs, the Capstone experience really stood out to me. I'm really looking forward to the hands-on experiences and opportunities to not only learn what companies are already doing but actually apply what we are learning in our courses to help find new solutions and**

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Full-Time | MS in Business Analytics Curriculum (9 months)

Download Curriculum Map [pdf]

Read Course Descriptions

Mini 1

Mini 2

Mini 3

Part-Time | MS in Business Analytics Curriculum (20 months)

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Yr 1 - Fall Mini 2

Yr 1 - Spring Mini 3

Yr 1 - Spring Mini 4

Yr 1 - Summer Mini 5

Yr 1 - Summer Mini 6

Yr 2 - Fall Mini 1

Yr 2 - Fall Mini 2

Yr 2 - Spring Mini 3

Yr 2 - Spring Mini 4

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Our students develop proficiency in the full range of state-of-the-art business analytics techniques; they also learn how to tell stories through and extract insights from data. Given the Tepper School's view of a curriculum as an organic entity, our faculty continually work in concert to ensure that courses harmonize, even as they are individually updated and modified to ensure learning outcomes for students are always in step with an ever-evolving industry.

Whether you enroll full-time on campus or in the flexible online format, students will make meaningful personal and professional connections with each other and develop the strong communication, collaboration and virtual team skills that will help them succeed in today's increasingly interconnected and global business environments.

Business Analytics Learning Outcomes

Upon completing the program, MSBA students will be able to:

- Demonstrate a depth of knowledge of quantitative and analytical tools for decision-making.
- Make appropriate judgments regarding managing, manipulating, and analyzing large data sets.
- Develop and/or efficiently apply computer software to implement analytical techniques.
- Identify the potential and challenges of applying data analytics in a business environment.
- Communicate effectively and persuasively.

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Master of Science in Business Analytics Courses

The Master of Science in Business Analytics (MSBA) courses equip students with leading-edge knowledge, skills, and experiential training in methodology, software engineering, corporate communication, and business domain knowledge.

46-880 Introduction to Probability and Statistics (6 units)

46-881 Programming in R and Python (6 units)

46-882 Business Fundamentals (6 units)

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46-886 Machine Learning Fundamentals (6 units)

46-887 Machine Learning for Business Applications (6 units)

46-888 Optimization (6 units)

46-889 Business Value for Integrative Analytics (6 units)

46-890 Managing People and Teams (6 units)

46-892 Data Analytics in Finance (6 units)

46-893 Operations and Supply Chain Analytics (6 units)

46-894 Analytical Marketing (6 units)

46-897 Business Communication (6 units)

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Connect with a Student Ambassador

Get firsthand insight into the Tepper School MSBA experience. Our student ambassadors are ready to share what life is like in the program and answer your questions.

Chat with a Student

About CMU

Athletics

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Top Tools

- Report Digital Accessibility Barrier
- Academic Calendar
- Bookstore
- Canvas
- The HUB
- Workday



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